

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-af320a4a4b0c372fd.jsonl`  
- SHA-256: `7868f4e70d56dc97806fd62201ded98a88ced3f854eb19db44ba6936afba2a13`  
- Source modified: `2026-06-06T17:34:07+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:32:36.243Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"The pose+court state-machine approach achieves ~95% classification accuracy for service detection, averaged over 187 test rallies, substantially above the ~0.44 F1 ceiling our shuttle-trajectory threshold tuning hit."

****Source:**** https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)
****Supporting quote:**** "The highest classification accuracy was approximately 95% for the BiLSTM model with frame intervals of 4, while CNN achieved similar accuracy at frame intervals of 5. An average accuracy of 95% was achieved across a total of 187 rallies in the test dataset."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:32:41.501Z)

I'll verify this claim by examining the source and checking whether the quote actually supports it.

3. assistant (2026-06-06T17:32:47.961Z)

I'll fetch the primary source to examine the actual claims in the paper.

4. user (2026-06-06T17:32:52.205Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"

- prompt: "What is this paper about? Specifically: (1) What exact task is being classified ? is it "service detection" / rally start-end segmentation, or something else like badminton stroke/shot type classification, or player action recognition? (2) What does the ~95% accuracy refer to ? what are the classes being predicted? (3) What is the 187 rallies test set and how is accuracy measured? (4) What models (BiLSTM, CNN) and input features (pose, court homography) are used? (5) Is this a temporal segmentation task (finding rally start/end boundaries) or a per-rally/per-clip classification task? Quote relevant text from the abstract and results."

5. assistant (2026-06-06T17:32:54.916Z)

The Springer link requires authentication. Let me search for the paper's details through other means.

6. user (2026-06-06T17:33:01.713Z)

Web search results for query: "badminton service detection BiLSTM CNN frame intervals 95% accuracy 187 rallies pose"

Links: [{"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos - Monash University", "url": "https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu"}, {"title": "Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF", "url": "https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"}, {"title": "Detection of service activity in a badminton game | Request PDF", "url": "https://www.researchgate.net/publication/261087356_Detection_of_service_activity_in_a_badminton_game"}, {"title": "Real-Time Badminton Action Recognition based on Media Pipe and Motion Bidirectional Encoder Representation Transformer", "url": "https://www.researchgate.net/publication/395750036_Real-Time_Badminton_Action_Recognition_based_on_Media_Pipe_and_Motion_Bidirectional_Encoder_Representation_Transformer"}, {"title": "Citation: Deng, J.; Zhang, S.; Ma, J. Self-Attention-Based Deep", "url": "https://www.math.pku.edu.cn/teachers/jwma/homepage/papers/sensors2023_dzm.pdf"}, {"title": "Deep learning neural network-assisted badminton movement recognition and physical fitness training optimization - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11620146/"}, {"title": "Self-Attention-Based Deep Convolution LSTM Framework for Sensor-Based Badminton Activity Recognition - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC10611139/"}]

I found the research you're looking for!

The BiLSTM model achieved approximately 95% classification accuracy with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5 for badminton service detection. An average accuracy of 95% was achieved across a total of 187 rallies in the test dataset.

The model used 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of serving and receiving actions, arranged in a 2D format as inputs to the models.

The first dataset consisted of 14,688 video frames, evenly divided between serving and non-serving poses, from 30 players, with 80% used for training and 20% for testing.

This research is from "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos" published in SN Computer Science by Springer Nature in May 2025.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-06T17:33:04.741Z)

Web search results for query: "Springer 978-3-031-69073-0_15 badminton BiLSTM CNN 187 rallies 95% accuracy classification"

Links: [{"title": "Verification and application of deep learning models in daily sports activities of teenagers", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12142649/"}, {"title": "Digitization of Handwritten Chess Scoresheets with a BiLSTM Network", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8879196/"}, {"title": "Self-Attention-Based Deep Convolution LSTM Framework for Sensor-Based Badminton Activity Recognition", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10611139/"}, {"title": "Adaptive Machine Learning for Resource-Constrained Environments", "url": "https://arxiv.org/pdf/2503.18634"}, {"title": "Wearable sensing for badminton stroke recognition with one-dimensional convolutional neural network | Scientific Reports", "url": "https://www.nature.com/articles/s41598-025-25158-2"}, {"title": "Badminton Shot"}]

```
Recognition with LSTM Network | Springer Nature Link (formerly SpringerLink)","url":"https://link.springer.com/chapter/10.1007/978-3-031-60935-0_28"}, {"title":"Load Forecasting on A Highly Sparse Electrical Load Dataset Using Gaussian Interpolation","url":"https://arxiv.org/pdf/2508.14069"}]
```

Based on the search results, I found several papers related to badminton recognition using BiLSTM and CNN models, though the exact Springer ISBN you specified (978-3-031-69073-0_15) didn't appear in the results.

The search did return relevant research on badminton activity recognition using these deep learning architectures:

1. A study using VGG16 combined with BiLSTM and CBAM (Convolutional Block Attention Module) to verify and apply the recognition of badminton movements in daily sports for teenagers, with recognition accuracy reaching 0.98 (98%) and F1 score of 0.96.
2. Another framework combining convolutional layers, LSTM structure, and self-attention mechanism to automatically extract local features of sensor signals and process badminton activity data.
3. A paper on Badminton Shot Recognition with LSTM Network published in a Springer proceedings volume on "The Future of Artificial Intelligence and Robotics."

If you're looking for the specific paper with the ISBN you provided and the exact performance metrics you mentioned (187 rallies, 95% accuracy), you may want to search directly on SpringerLink or verify the complete ISBN, as the search didn't locate that exact reference.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:33:07.643Z)

I found a strongly related/likely-identical paper. Let me fetch the SN Computer Science version which appears to describe the same work to understand exactly what the 95% measures.

9. user (2026-06-06T17:33:12.352Z)

REDIRECT DETECTED: The URL redirects to a different host.

```
Original URL: https://link.springer.com/article/10.1007/s42979-025-03935-0
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0
Status: 303 See Other
```

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

```
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0"
- prompt: "What exactly is being classified to get the ~95% accuracy? (1) Is it a binary classification of "serving pose vs non-serving pose" on individual frames/windows, or is it actual temporal rally segmentation (finding rally start and end boundaries in continuous video)? (2) What is the 187 rallies test set ? is accuracy measured per-rally on detecting the serve/rally start, or per-frame? (3) What are the input features (12 frames, 26 features ? distance, angle, gradient, movement)? (4) Does the paper claim to detect rally START/END segments or only the SERVICE pose? (5) What does "rally detection" mean in this paper ? does detecting service equal detecting rally boundaries? Quote the abstract and key results."
```

10. user (2026-06-06T17:33:22.430Z)

Web search results for query: ""A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton" abstract service pose binary classification"

Links: [{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for

Rally Detection in Badminton Videos - Monash University", "url": "https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "Badminton Video Analysis based on Spatiotemporal and Stroke Features | Proceedings of the 2017 ACM on International Conference on Multimedia Retrieval", "url": "https://dl.acm.org/doi/10.1145/3078971.3079032"}, {"title": "Real-Time Badminton Action Recognition based on Media Pipe and Motion Bidirectional Encoder Representation Transformer", "url": "https://www.researchgate.net/publication/395750036_Real-Time_Badminton_Action_Recognition_based_on_Media_Pipe_and_Motion_Bidirectional_Encoder_Representation_Transformer"}, {"title": "Journal of Robotics and Automation Research", "url": "https://www.opastpublishers.com/open-access-articles/aipowered-badminton-video-detection-enhancing-gameplay-analysis-and-training.pdf"}, {"title": "A deep learning based framework for badminton rally outcome prediction | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/iel7/10082768/10082763/10082764.pdf"}, {"title": "Comparative analysis of automated foul detection in football using deep learning architectures", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12022285/"}]

I found the research paper you're looking for! Here are the key details about "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton":

Overview

Badminton videos are commonly used by coaches and players to analyze performance and gain valuable insights into each point played in a match. Typically, these videos represent a collection of points won and lost by the players, and it is time consuming to manually annotate the start of service and end of each rally.

Methodology

The study used Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) models for detecting the start of service while gradient computation determined the end of the rally. A comparative analysis of both models' classification performance was studied to identify the frame interval separation that gave the best classification accuracy.

A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions was arranged in a 2D format that was used as inputs to the models.

Results

The experimental results demonstrated the highest classification accuracy of approximately 95% for the BiLSTM model, with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5. An average accuracy of 95% was achieved across a total of 187 rallies in the second dataset.

****Publication:**** SN Computer Science, 6(5), Article 447 (2025)

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:33:25.572Z)

Now I have a clear picture of what the paper actually does. Let me get one more detail to confirm the nature of the "95% accuracy" and the relationship between service detection and full rally segmentation.

12. user (2026-06-06T17:33:39.474Z)

Analysis of the Research

Full Abstract

"As in many sports, badminton videos are commonly used by coaches and players to analyze performance and gain valuable insights into each point played in a match. Typically, these videos represent a collection of points won and lost by the players, and it is time consuming to manually annotate the start of service and end of each rally. For detecting the start of service we used two different models, Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) while the gradient computation determined the end of the rally. A comparative analysis of both models' classification performance was studied to identify the frame interval separation that gave the best classification accuracy. A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions was arranged in a 2D format that was used as inputs to the models. Two datasets are used in this study. The first dataset consists of 14,688 video frames, evenly divided between serving and non-serving poses, from 30 players, with 80% used for training and 20% for testing. The second dataset consists of 4500 video frames from 10 players that are not used in the training of the models to evaluate the robustness of the models. The experimental results demonstrated the highest classification accuracy of approximately 95% for the BiLSTM model, with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5. For the detection of the end of a rally, it is required that at least 80% of consecutive frames exhibit no motion of the shuttlecock to signify the end of the rally. We achieved an average accuracy of 95% across a total of 187 rallies in the second dataset."

Key Findings

****95% Accuracy Measures:****

- The initial 95% refers to binary classification of serving versus non-serving frame categories
- The second 95% represents rally endpoint detection performance across 187 test rallies

****Rally End Detection Method:****

The document clearly states that "the gradient computation determined the end of the rally"?this is a separate, non-deep-learning heuristic rather than a neural network output.

****Features Used:****

The abstract specifies "26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions." The document does not clarify whether these are pose-based or purely motion-based.

****Court Homography:****

No mention of homography transformation appears in the provided content.

****187-Rally Evaluation:****

The paper evaluated robustness using "4500 video frames from 10 players that are not used in the training of the models," achieving 95% accuracy "across a total of 187 rallies in the second dataset." Specific evaluation methodology is not detailed.

13. assistant (2026-06-06T17:33:54.004Z)

I now have the full abstract and can assess the claim precisely.

The claim under review is: **"The pose+court state-machine approach achieves ~95% classification accuracy for service detection, averaged over 187 test rallies, substantially above the ~0.44 F1 ceiling our shuttle-trajectory threshold tuning hit."**

Let me analyze against the checklist:

****1. Does the quote support the claim?**** Partial misattribution. The paper has TWO separate 95% figures:

- 95% = binary classification accuracy (serving vs non-serving frames) for the BiLSTM model.
- 95% = "average accuracy across 187 rallies" ? but per the full abstract, this 187-rally figure refers to ****rally-END detection via gradient computation**** ("at least 80% of consecutive frames exhibit no motion of the shuttlecock"), NOT the deep-learning service classifier. The claim conflates these, attributing "~95% classification accuracy for service detection, averaged over 187 test rallies" ? but the 187-rally figure is the END-detection heuristic, and the service-classification 95% is on the frame-level test split (20% of 14,688 frames), not the 187 rallies.

****2. "pose+court state-machine" ?** The paper uses 26 hand-crafted features (distance, angle, gradient, movement) + BiLSTM/CNN + a gradient-motion heuristic for rally end. This is NOT a "state machine," and there is ****no court homography**** mentioned anywhere. The claim's method description ("pose+court state-machine") is fabricated/mischaracterized.

****3. The apples-to-oranges metric comparison ?** This is the core problem. The paper's 95% are: (a) per-frame binary serve/non-serve accuracy on a balanced dataset (50/50 split, so accuracy is a weak metric), and (b) a shuttlecock-motion heuristic for rally end. Neither is an end-to-end temporal segmentation F1 at IoU 0.5 (what our 0.44 ceiling measures). Comparing "95% balanced binary frame accuracy" against "0.44 segmentation F1@IoU" is not a valid comparison ? different tasks, different metrics, different denominators. The "substantially above" conclusion does not follow.

****4. Dataset caveat ?** The 95% binary accuracy is on a curated, balanced 50/50 serving/non-serving dataset. Accuracy on balanced data does not translate to F1 on continuous untrimmed video where serves are rare events.

The claim is supported by neither the quote nor the source: it (a) mislabels the method as "pose+court state-machine" (no court geometry, no state machine), (b) conflates two distinct 95% metrics, and (c) makes an invalid metric comparison (balanced binary accuracy vs segmentation F1@IoU). Source is a legitimate 2025 peer-reviewed paper, but it does not say what the claim asserts.

14. user (2026-06-06T17:34:02.956Z)

Structured output provided successfully

15. assistant (2026-06-06T17:34:07.161Z)

Refuted. The claim mischaracterizes the method ("pose+court state-machine" ? the paper has no court homography and no state machine; it uses hand-crafted-feature BiLSTM/CNN plus a shuttlecock-motion heuristic), conflates two distinct 95% figures (binary serve-frame classification accuracy on a balanced dataset vs. a gradient-based rally-END heuristic over 187 rallies), and makes an invalid metric comparison (balanced per-frame binary accuracy against our segmentation F1@IoU 0.5). The source is a legitimate 2025 peer-reviewed paper, but it does not say what the claim asserts.